=== UNIT 5: MULTIVARIATE INTEGRAL CALCULUS (DOUBLE & TRIPLE INTEGRALS) ===

>> 5.1 Double Integrals and Change of Order of Integration

Multivariate integration extends single-variable integration to planar domains and 3D spatial volumes.
* Double Integral over Region R: iint_R f(x, y) dA = int_{x=a}^b [int_{y=phi1(x)}^{phi2(x)} f(x, y) dy] dx.
* Change of Order of Integration: When an iterated integral int_a^b [int_{y1}^{y2} f(x, y) dy] dx is difficult or impossible to evaluate analytically in its given order, changing the integration order to dx dy often renders the antiderivative elementary. Requires sketching the boundary curves of region R, determining whether slices are vertical (dx first) or horizontal (dy first), and identifying new continuous limit functions.

>> 5.2 Transformation of Coordinates in Multiple Integrals

* General 2D Coordinate Transformation (x = x(u, v), y = y(u, v)): iint_R f(x, y) dx dy = iint_{R'} f(x(u, v), y(u, v)) * | J | du dv, where | J | = | d(x, y) / d(u, v) | is the absolute value of the transformation Jacobian.
* Polar Transformation: x = r*cos(theta), y = r*sin(theta), dx dy = r dr d(theta). Essential for circular, annular, or radial boundaries.
* Triple Integrals in Cartesian Coordinates: iiint_V f(x, y, z) dx dy dz.
* Cylindrical Coordinates: x = r*cos(theta), y = r*sin(theta), z = z. Volume element dV = r dr d(theta) dz.
* Spherical Polar Coordinates: x = rho*sin(phi)*cos(theta), y = rho*sin(phi)*sin(theta), z = rho*cos(phi). Volume element dV = rho^2 * sin(phi) d(rho) d(phi) d(theta).

>> 5.3 Geometric Applications: Area, Volume and Center of Mass

* Planar Area: Area(R) = iint_R dx dy = iint_R r dr d(theta).
* Spatial Volume: Volume(V) = iiint_V dx dy dz.
* Mass and Center of Mass: Mass M = iint_R sigma(x, y) dA. Coordinates of centroid: x_bar = (1/M) * iint_R x*sigma dA, y_bar = (1/M) * iint_R y*sigma dA.

[SOLVED PROBLEM 1: Changing Order of Integration of Non-Elementary Antiderivative]

Problem: Evaluate the integral I = int_0^1 [int_x^1 e^{y^2} dy] dx by reversing the

order of integration.
Solution Step 1: Identify given region R from integration limits:
Inner integral: y runs from y = x to y = 1.
Outer integral: x runs from x = 0 to x = 1.
Region R is the triangle bounded by the line y = x, the horizontal line y = 1, and the y-axis (x = 0).
Solution Step 2: Reverse order to integrate with respect to x first, then y:
For a fixed horizontal slice y, x varies from x = 0 (y-axis) to x = y (the boundary line).
y varies along the full vertical height of the region from y = 0 to y = 1.
Transformed Integral: I = int_{y=0}^1 [int_{x=0}^y e^{y^2} dx] dy.
Solution Step 3: Evaluate inner integral with respect to x (treating e^{y^2} as constant):
int_0^y e^{y^2} dx = e^{y^2} * [x]_0^y = y * e^{y^2}.
Solution Step 4: Evaluate outer integral with respect to y:
I = int_0^1 y * e^{y^2} dy.
Let u = y^2 => du = 2y dy => y dy = du / 2.
When y = 0, u = 0; when y = 1, u = 1.
I = (1 / 2) * int_0^1 e^u du = (1 / 2) * [e^u]_0^1 = (1 / 2) * (e^1 - e^0) = (e - 1) / 2.
Final Numerical Value: I = (2.71828 - 1) / 2 = 0.85914.

[SOLVED PROBLEM 2: Double Integral over Circular Domain via Polar Coordinates]

Problem: Evaluate iint_R sqrt(a^2 - x^2 - y^2) dx dy over the positive first quadrant of the circle x^2 + y^2 <= a^2.
Solution Step 1: Convert to polar coordinates: x = r*cos(theta), y = r*sin(theta), dx dy = r dr d(theta).
Integrand: sqrt(a^2 - (x^2 + y^2)) = sqrt(a^2 - r^2).
Solution Step 2: Determine limits for the first quadrant of circle of radius a:
r runs from r = 0 to r = a.
theta runs from theta = 0 to theta = pi/2.
Solution Step 3: Set up and evaluate iterated integral:
I = int_{theta=0}^{pi/2} [int_{r=0}^a sqrt(a^2 - r^2) * r dr] d(theta).
Inner Integral: Let u = a^2 - r^2 => du = -2r dr => r dr = -du / 2.
When r = 0, u = a^2; when r = a, u = 0.
int_{a^2}^0 sqrt(u) * (-du / 2) = (1 / 2) * int_0^{a^2} u^{1/2} du = (1 / 2) * [(2/3) * u^{3/2}]_0^{a^2}
= (1 / 3) * (a^2)^{3/2} = a^3 / 3.
Solution Step 4: Outer Integral:
I = int_0^{pi/2} [a^3 / 3] d(theta) = (a^3 / 3) * [theta]_0^{pi/2} = (a^3 / 3) * (pi / 2) = pi * a^3 / 6.
Final Answer: Integral value = pi * a^3 / 6.

[SOLVED PROBLEM 3: Triple Integral Volume of Sphere via Spherical Coordinates]

Problem: Evaluate the volume of a sphere of radius R using spherical polar coordinates.
Solution Step 1: Coordinate transformation: x = rho*sin(phi)*cos(theta), y = rho*sin(phi)*sin(theta), z = rho*cos(phi).
Jacobian volume element: dV = rho^2 * sin(phi) d(rho) d(phi) d(theta).
Solution Step 2: Define spherical limits for full sphere:
Radial distance rho: 0 to R.
Colatitude (polar angle) phi: 0 to pi.
Azimuthal angle theta: 0 to 2*pi.
Solution Step 3: Set up and separate independent triple integral:
Volume V = int_0^{2*pi} d(theta) * int_0^pi sin(phi) d(phi) * int_0^R rho^2 d(rho).
Integral 1: int_0^{2*pi} d(theta) = 2*pi.
Integral 2: int_0^pi sin(phi) d(phi) = [-cos(phi)]_0^pi = -cos(pi) - (-cos(0)) = -(-1) + 1 = 2.
Integral 3: int_0^R rho^2 d(rho) = [rho^3 / 3]_0^R = R^3 / 3.
Solution Step 4: Multiply results:
V = (2*pi) * (2) * (R^3 / 3) = (4/3) * pi * R^3. (Exact Standard Formula Proven).

[SOLVED PROBLEM 4: Centroid of Homogeneous Semi-Circular Laminar Disc]

Problem: Find the center of mass (x_bar, y_bar) of a uniform thin semi-circular plate of radius a lying in the upper half plane x^2 + y^2 <= a^2, y >= 0.
Solution Step 1: Symmetry consideration: The laminar disc is symmetric about the y-axis (x = 0).
Therefore, x_bar = 0.
Solution Step 2: Total area of semi-circle: Area A = pi * a^2 / 2.
Solution Step 3: Compute first moment of area M_x = iint_R y dA in polar coordinates:
y = r*sin(theta), dA = r dr d(theta).
M_x = int_0^pi [int_0^a (r*sin(theta)) * r dr] d(theta) = int_0^pi sin(theta) d(theta) * int_0^a r^2 dr.
Angular integral: int_0^pi sin(theta) d(theta) = [-cos(theta)]_0^pi = 1 - (-1) = 2.
Radial integral: int_0^a r^2 dr = [r^3 / 3]_0^a = a^3 / 3.
M_x = 2 * (a^3 / 3) = 2*a^3 / 3.
Solution Step 4: Calculate centroid coordinate y_bar:
y_bar = M_x / Area = [2*a^3 / 3] / [pi * a^2 / 2] = (2*a^3 / 3) * (2 / (pi * a^2)) = 4*a / (3*pi).
Final Centroid Coordinates: (x_bar, y_bar) = (0, 4a / (3*pi)). For a = 3 m, y_bar ~ 1.273 m.

>> 5.4 University Review & Examination Questions

* Q1: State the general formula for transforming variables in a double integral and explain the physical significance of the Jacobian determinant.
Answer: Formula: iint_R f(x, y) dx dy = iint_{R'} f(x(u, v), y(u, v)) * |J| du dv, where J = d(x, y)/d(u, v). Physical significance: The Jacobian determinant represents the local magnification or contraction ratio of area elements when mapping from the (u, v) plane to the (x, y) plane. An infinitesimal rectangular area du*dv in the parameter space maps to an infinitesimal parallelogram in the Cartesian plane spanned by vectors (dx/du du, dy/du du) and (dx/dv dv, dy/dv dv). The area of this parallelogram equals the cross product magnitude, which evaluates precisely to |(dx/du)(dy/dv) - (dx/dv)(dy/du)| * du dv = |J| du dv.
* Q2: Explain the procedure for changing the order of integration in double integrals with an illustrative sketch description.
Answer: Procedure: (1) Identify given boundary curves from integration limits. (2) Accurately sketch the 2D domain R enclosed by these curves. (3) If the original integral integrated with respect to y first (vertical strip extending from y1(x) to y2(x)), convert to horizontal strips by finding x as a function of y (x1(y) to x2(y)). (4) Determine the global minimum and maximum bounds of y across the domain (c to d). (5) If the boundary equations change across the region, partition R into sub-regions and sum the resulting iterated integrals.

>> 5.5 Key Takeaways and Revision Points

* Reversing the order of integration converts analytically impossible single-direction antiderivatives into elementary integrals.
* Polar transformation (dx dy = r dr d(theta)) simplifies circular and radial domain integrations.
* Triple integrals evaluate 3D spatial properties (volume, mass, gravitational potential).
* Spherical coordinates volume element is dV = rho^2 * sin(phi) d(rho) d(phi) d(theta).